

LinReTraCe – Installation & Tutorial

Matthias Pickem 2024

System requirements:

Modern GNU/Linux Distribution (Linux Kernel $\geq 4.x$ + Python3, e.g., Ubuntu 24.04 LTS)

Fortran Compiler with quadruple precision support

System prerequisites: pip, git, make, cmake, Fortran95 compiler (e.g., gfortran), C and C++ compiler (e.g., gcc and g++)

Packages are obtainable, e.g., under Ubuntu, via `sudo apt install pip`.

You need admin rights to do so, or you have to ask your computer admin.

Pip can be also obtained via the `get-pip.py` script, circumventing some of the root requirements. (<https://pip.pypa.io/en/stable/installation/#get-pip-py>)

Modern systems use virtual environments for package installations. The longer installation procedure comes with the benefit that installations of other programmes/packages will never interfere with LinReTraCe. For Ubuntu 24.04 LTS the use of an environment management is enforced.

Recommended installation procedure using anaconda

LinReTraCe installation from a clean Ubuntu 24.04 LTS installation

- 1) Installation of standard packages
`sudo apt install git pip gfortran gcc g++ cmake`
- 2) Installation of anaconda
`curl https://repo.anaconda.com/archive/Anaconda3-2024.02-1-Linux-x86_64.sh > Anaconda3-2024.02-1-Linux-x86_64.sh # command in 1 line!`
`bash Anaconda3-2024.02-1-Linux-x86_64.sh`
This activates anaconda's 'base' environment. Additional environments can now be managed via "conda env"
- 3) Setting up of a conda environment "lrtc" for LinReTraCe
`conda create --name lrtc python=3.11`
`conda activate lrtc # the environment needs to be activated whenever you use LinReTraCe`
If needed later, you can use `conda activate base` to switch back to the default environment. The restriction to python version 3.11 (see above) is currently needed for the optional `bolztrap2` package (and its `pyfftw` dependency) to work.
- 4) Installation of the hdf5 library
Option 1: Use of version 1.13.3
`export HDF5SOURCE=hdf5-1.13.3`
`mkdir ~/libs`
`wget`
<https://support.hdfgroup.org/ftp/HDF5/releases/hdf5-1.13/hdf5-1.13.3/src/hdf5-1.13.3.tar.gz> `-P ~/libs/`

Option 2: for the latest version of the hdf5 library go to
<https://www.hdfgroup.org/downloads/hdf5/source-code/>

login/register (for free!) and download the hdf5 tar-file, e.g., hdf5-1.14.4-2.tar.gz

```
export HDF5SOURCE=hdf5-1.14.4-2 # adapt version number here!
```

```
mkdir ~/libs
```

```
mv ~/Downloads/$HDF5SOURCE.tar.gz ~/libs
```

For both options continue with

```
cd ~/libs
```

```
tar xf $HDF5SOURCE.tar.gz
```

```
cd $HDF5SOURCE
```

```
CC=gcc
```

```
FC=gfortran
```

```
mkdir ~/opt
```

```
./configure --enable-fortran --prefix=/home/$USER/opt/"$HDF5SOURCE"_gcc
```

```
make
```

```
make check
```

```
make install
```

```
export
```

```
LD_LIBRARY_PATH=/home/$USER/opt/"$HDF5SOURCE"_gcc/lib:$LD_LIBRARY_PATH
```

In order not to do this every time, let's add this to the .bashrc

```
echo 'export
```

```
LD_LIBRARY_PATH=/home/$USER/opt/"$HDF5SOURCE"_gcc/lib:$LD_LIBRARY_PATH' >> ~/.bashrc
```

- 5) Download the LinReTraCe code from github (a subfolder "linretrace" will be created)

```
mkdir ~/programs # or adapt to your program folder
```

```
cd ~/programs
```

```
git clone https://github.com/linretrace/linretrace
```

- 6) Install required python packages

```
cd linretrace
```

```
pip install -e . or pip install --use-pep517 -e . in case later on a pkg error occurs
```

```
pip install cmake packaging pyfftw boltztrap2 # if you require boltztrap2 features
```

- 7) save the following into make_include. If necessary, adapt the version number of the hdf5 library (see folder make_include_examples/ for examples, e.g., for a compilation with parallel computing (MPI) capabilities)

```
FC = gfortran
```

```
FCDG = gfortran
```

```
FFLAGS = -O3
```

```
HDF5 = -I/home/$USER/opt/hdf5-1.14.4-2_gcc/include
```

```
HDF5 += -L/home/$USER/opt/hdf5-1.14.4-2_gcc/lib -lhdf5_fortran -lhdf5hl_fortran
```

- 8)

```
make validate
```

```
make linretrace
```

```
make install
```

- 9)

```
python setup.py install
```

Installation complete. The linretrace binary will be located in \$HOME/bin. The python script ltb, lprint, ... will be located in a subfolder of the anaconda environment, e.g., /home/\$USER/libs/anaconda3/envs/lrtc/bin/lbt. All executables will be globally accessible.

Installation procedure using pip alone

If one only has access to the in-built python installation, the way of doing this looks as follows: Inside the package execute

- **python -m venv myenv** : create localized environment
- **source myenv/lib/activate** : activate the environment

This creates a subfolder myenv, into which a stand-alone python + pip package is loaded into. Any execution of “python” or “pip” thus references this localized environment, which packages are installed into and loaded from. Leaving the environment is done by typing “deactivate”.

This “**venv workflow**” has been confirmed to work for **Python 3.9** (or older) if the optional Boltztrap package is required. LinReTraCe itself works with the latest versions (currently Python 3.12).

Installation without environment (will not work on Ubuntu 24.04 LTS)

LinReTraCe installation from a clean Ubuntu 22.04 LTS installation:

- 1) `sudo apt install git pip gfortran gcc g++ cmake`
- 2) `pip install numpy scipy h5py matplotlib ase spglib`
- 3) `pip install cmake packaging pyfftw boltztrap2`
- 4) **HDF5 installation** (USER=your username)
 - `cd` # or choose a custom directory where to install the library
 - `mkdir -p opt`
 - `wget`
<https://support.hdfgroup.org/ftp/HDF5/releases/hdf5-1.13/hdf5-1.13.3/src/hdf5-1.13.3.tar.gz>
 - `tar xf hdf5-1.13.3.tar.gz`
 - `cd hdf5-1.13.3`
 - `CC=gcc`
 - `FC=gfortran`
 - `./configure --enable-fortran --prefix=/home/$USER/opt/hdf5-1.13.3_gcc`
 - `make`
 - `make install`
- 5) **LinReTraCe installation**
 - `cd` # or choose a custom directory where to install the code
 - `git clone https://github.com/linretrace/linretrace`
 - `cd linretrace`
 - with root rights: `sudo python3 setup.py install`
 - without root rights: `python3 setup.py install --user`
 - `touch make_include`
 - save the following into `make_include` (see folder `make_include_examples/` for examples, e.g., for a compilation with parallel computing (MPI) capabilities)

`FC = gfortran`

`FCDG = gfortran`

`FFLAGS = -O3`

`HDF5 = -I/home/$$USER/opt/hdf5-1.13.3_gcc/include`

`HDF5 += -L/home/$$USER/opt/hdf5-1.13.3_gcc/lib -lhdf5_fortran -lhdf5hl_fortran`

- `make validate`
- `make linretrace`
- `make install`
- `echo 'export PATH=/home/$USER/bin:$PATH' >> ~/.bashrc`
- `echo 'export`
`LD_LIBRARY_PATH=/home/$USER/opt/hdf5-1.13.3_gcc/lib:$LD_LIBRARY_PATH'`
`>> ~/.bashrc`
- `source ~/.bashrc`

Installation complete.

The linretrace binary will be located in \$HOME/bin.

The Python3 packages will be located either in
`/usr/local/lib/pythonX.Y/site-packages`
or
`$HOME/.local/lib/pythonX.Y/site-packages`
depending on how `setup.py` was called.
(see <https://docs.python.org/3/install/> for a detailed discussion)

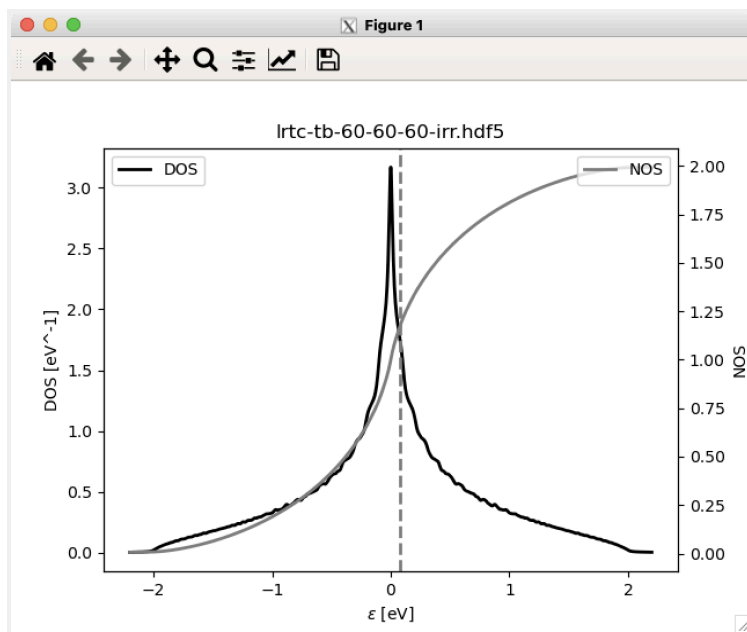
First LinReTraCe calculation: body centered-cubic tight-binding calculation

Setting up the energy file:

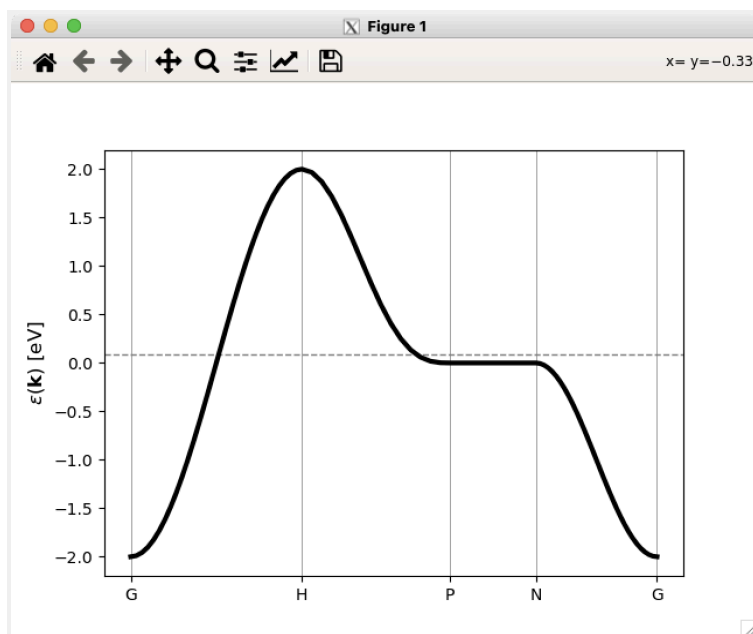
- `cd`
- `mkdir test`
- `cd test`
- `lrb ~/linretrace/templates/bcc.tbdata 60 60 60 1.2`

Inspecting energy file:

- `lprint lrtc-tb-60-60-60-irr.hdf5 dos -p` to plot the density of states:



- `lprint lrtc-tb-60-60-60-irr.hdf5 path GHPNG -p` to plot the band structure:



Setting up config file:

- lconfig

Starts up the following interactive prompt: Enter the highlighted inputs:

INTERACTIVE GENERATION OF LRTC CONFIG FILES

energy file: lrtc-tb-60-60-60-irr.hdf5

output file: output.hdf5

calculate intra-band quantities [y,n]: y

calculate inter-band quantities [y,n]: n

calculate boltzmann quantities [y,n]: y

calculate magnetic field quantities [y,n]: y

calculate quadruple precision [y,n]: y

run mode (temp [1], mu [2]): 1

use Scattering file [y,n]: n

TEMP-MODE configuration:

chemical potential (fermi [1], digamma [2]): 1

temperature starting point [K]: 100

temperature ending point [K]: 500

temperature points: 20

logarithmic temperature steps [y,n]: n

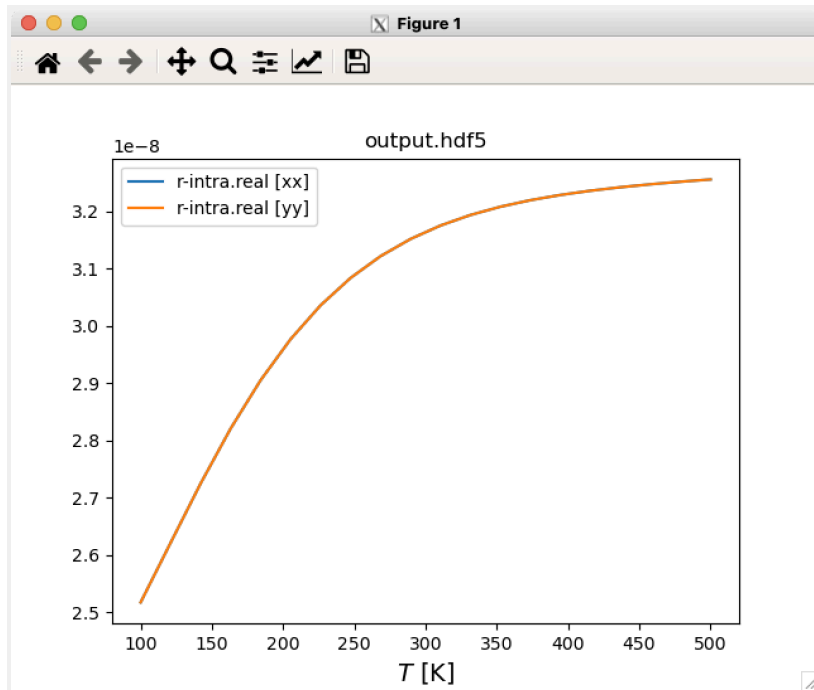
scattering coefficients (separated by space): 1e-2

quasiparticle coefficients (separated by space): 1

config.lrtc successfully created.

Running LinReTraCe and inspecting output data:

- `linretrace config.lrtc` to execute the program
- `lprint -p latest r-intra xx yy` to plot the resistivity



- `lprint -p latest n-intra xyz` to plot the Nernst coefficient

